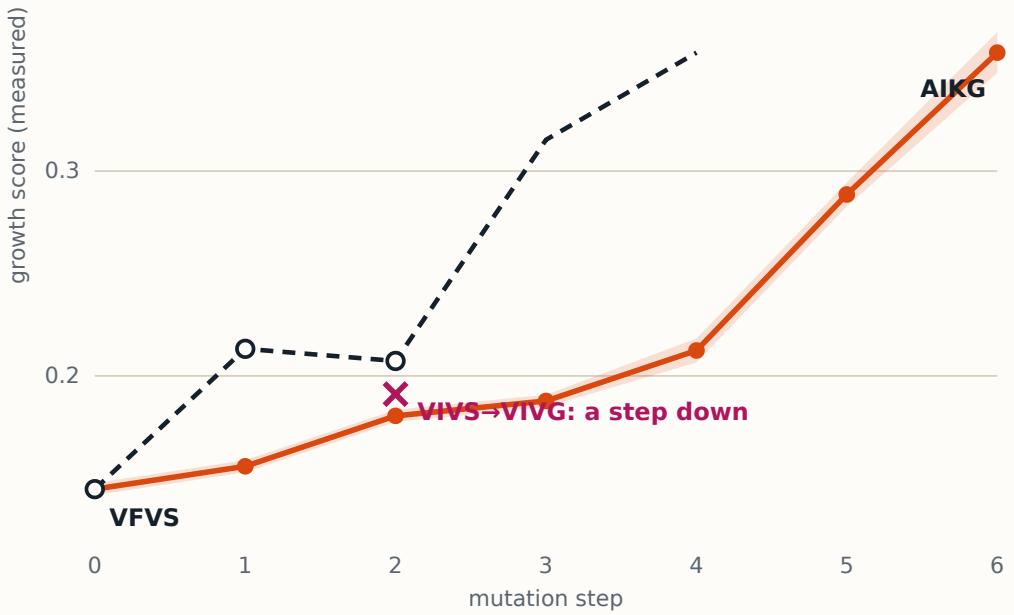
A protein design is not a plan.

Which mutation do you make first — and what evidence says the next step will still work?

TARGET AND THE MISSING ROUTE

TrpB enzyme, four residues, parent VFVS → best measured variant AIKG. All four changes are needed, so there are **24** orders to make them in. Taking the measurements at face value, **every one** of the 24 steps down at least once.

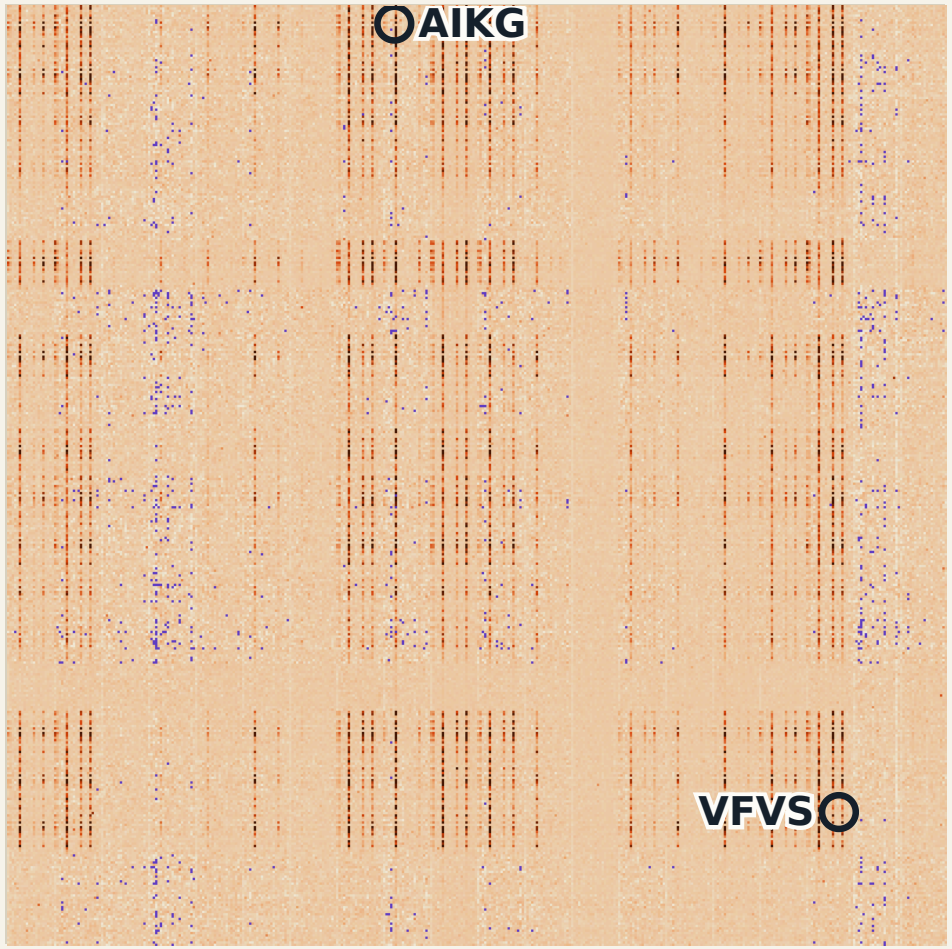


--- the most tempting direct ordering VFVS → VIVS → VIVG → VIKG → AIKG

— the 6-step detour whose every step is certified (band = assumed κ=2 interval)

Source Johnston et al., PNAS 2024, doi 10.1073/pnas.2400439121; data CaltechDATA 10.22002/h5rah-5z170 (CC0), dataset summary_AA.csv sha256 a631d8cf6c5b… · descriptive full-label figure: orderings recounted from site/data/landscape.js, cross-checked with artifacts/evidence/route_summary.json · artifacts @ a64c7210

This is a measured landscape: 159,129 of 160,000 variants have a laboratory value.



159,129
measured genotypes, two replicate flasks each

871
never measured — stay unknown

ROW LEDGER	COUNT
rows in source file (incl. stop codons)	193,170
stop-codon rows, excluded	34,041
measured, no stop	159,129
imputed in the authors' separate file	871
imputed rows used as observations here	**0**

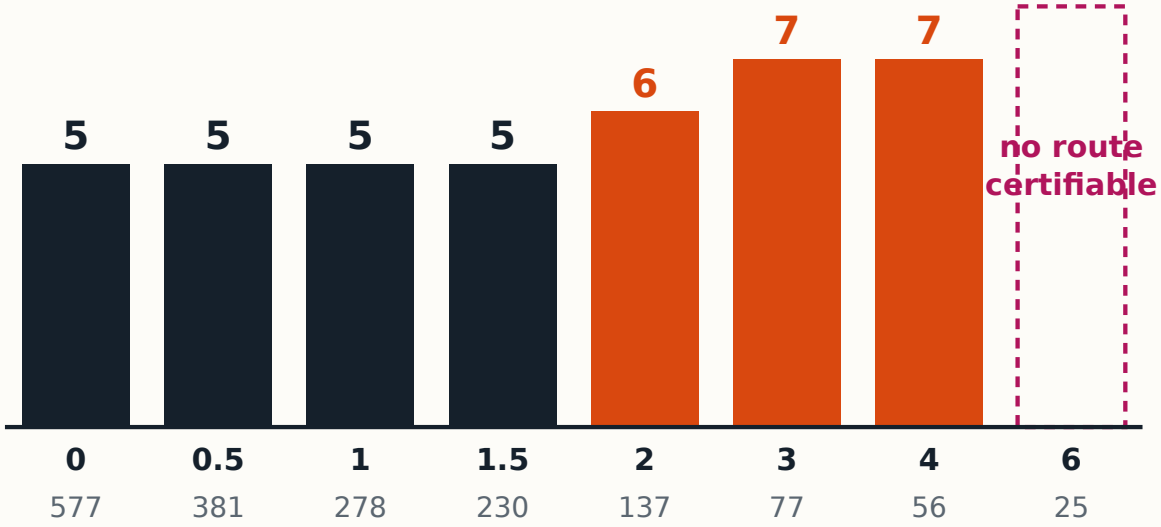
Colour = measured growth score, monotone gamma-stretched so low values are visible (cream low → deep orange high). The 871 unmeasured cells are violet and too small to see: each is a place a route could pass that we have no data for.

Reads as: cellular growth of E. coli selecting for tryptophan synthase activity — a proxy, not a purified-enzyme constant. Layout: rows = residues 1×2, columns = residues 3×4. A projection of sequence space, not molecular geometry.

Source dataset summary_AA.csv sha256 a631d8cf6c5b…; counts from artifacts/demo.json source.counts (independent recount: scripts/audit_data.py) · score = mean of two replicates' growth rate minus stop-codon background; not max-normalised · artifacts @ a64c7210

Challenge the assumption and the route moves: 5 steps → 6 → 7, then no route can be certified.

mutations on the shortest route whose every step is certified



bold row: assumed κ · grey row: variants reachable from VFVS by certified steps

κ multiplies the replicate-disagreement spread to widen every measurement into an interval. It is an **assumed** sensitivity — not a confidence level. A step is certified only if the next variant's lower bound clears the current one's upper bound.

The weakest step of the κ=2 route

SFMS → SIMS clears the bar by only **0.0013** growth units; the route is certifiable up to κ≈**2.45** and must be re-planned beyond.

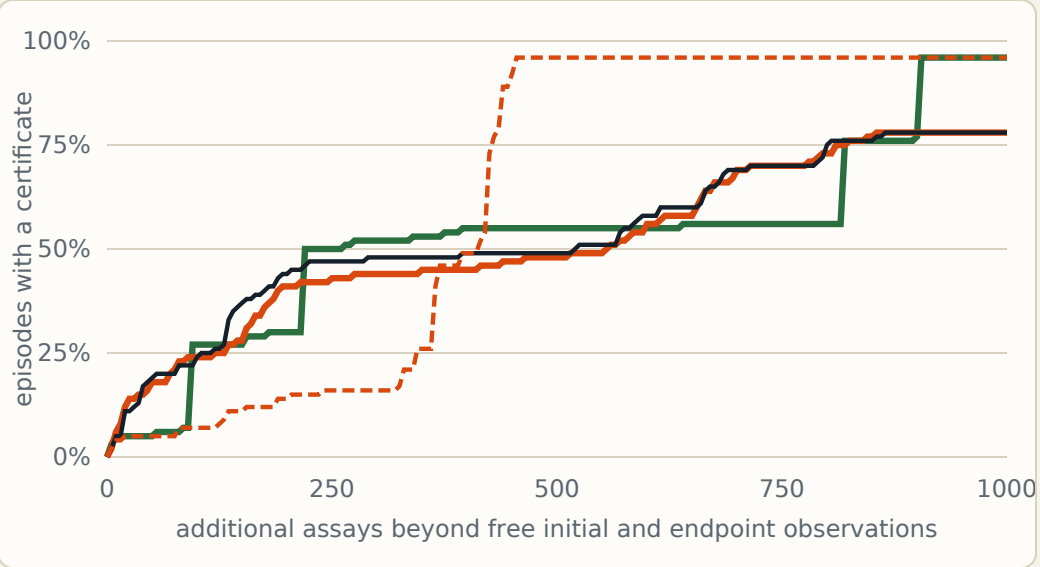
Variants reachable from VFVS by certified steps: **577** at κ=0 → **137** at κ=2 → **25** at κ=6. In the live tool, clicking a step shows its raw replicate rows.

Precisely: at κ=6 the tool says "no route certifiable at this bound" (25 variants reachable by certified steps). That does *not* mean no ascending route exists: overlapping intervals are simply not certified. κ=4 is the last bound in the sweep with a certified route.

When the evidence cannot settle a step, the system's output is the next assay to run — not a confidence number it cannot support.

Source artifacts/demo.json kappa_sweep (core worker; frozen noise model artifacts/core/protocol.json), independently recomputed at all 8 κ by scripts/independent_truth.py · descriptive, uses all measured labels (not a replay result); graph scope: measured 4-site subgraph, 871 unmeasured genotypes outside · artifacts @ a64c7210

The benchmark changed our design: established search certifies 96/100; our model-assisted planner 78/100.



POLICY	BASIS	@250	@1000	ASSAYS†
LazySP-style, alternate	established search · follow-up primary	50%	**96%**	477
query-to-certify (single)	model-assisted · original	43%	**78%**	483
specialists, shared	shared specialists · original	47%	**78%**	467
single planner, no GP (post hoc)	*POST HOC, chosen after seeing the original run*	*16%*	***96%***	*378*

Keep the evidence layer. Choose the simpler planner. No collective uplift. No superiority claim for Q2C. The LazySP adaptation is established work, not an invention.

LazySP-style alternate minus query-to-certify, paired by episode (percentage points; 95% interval; descriptive):

MEASURE	DIFFERENCE
certified @1000, preregistered seed bootstrap	**+18 [+14, +22]**
same, post hoc, 24 problem clusters	**+18 [+0, +51]**
mean assays to certificate (restricted)	**-6.5 [-53, +49]**
certified only by LazySP / only by Q2C	**18 / 0**, 17 of 18 at VFVS→ALKG

One enzyme; retrospective; assumed intervals. The gain concentrates in one repeated endpoint pair, and the assay-count interval includes 0. About 24 distinct problems, not 100 independent targets.

The follow-up was declared before its runs, after the original study. 0 false certificates among 883 issued checks structural consistency, not biology. † Unresolved counted at budget. All arms and limits are in the appendix.

Source artifacts/lazysp/results/ (3 variants × 100 episodes; 300 shards audited, 0 errors) · artifacts/core/replay/ (20 seeds × 5 pairs × 11 arms) · **Prior art** LazySP, BAEE, ALDE (not run here) · artifacts @ a64c7210

An existing Delphi claim now changes a decision: it runs when its assumptions hold and refuses when they don't.

Delphi claim
kimura-fixation-probability · Fisher FTNS
`content sha bbfd686b…`

→

hand-authored adapter
v1.0.0, pinned to the record
`src sha f371ae67…`

→

applicability guard
model, ploidy, quantity, inference computes or refuses

Provenance kept in four layers: source record · adapter mapping · new operationalization · result kind (**SIMULATED**, never observed).

REQUEST	DECISION	REASON
Kimura: diploid, genic, constant Ne (declared s)	COMPUTE	ok
Kimura on haploid E. coli	REFUSE	model_transfer_refused
Kimura with a TrpB growth score as s	REFUSE	unsupported_quantity_conversion
Fisher: partial change, declared V_A	COMPUTE	ok
Fisher: total fitness change	REFUSE	inference_refused
Fisher: "every step improves"	REFUSE	inference_refused

Third example, model-derived: an existing Delphi Turing record on a finite Neumann domain has no admissible unstable mode at L=π but one (n=1, determinant -1/8) at L=2π. Lean checks the dispersion algebra, not the PDE or any biology.

SAME REQUEST, DIFFERENT SOURCE STATE

source withdrawn	compute → REFUSE	claim_withdrawn
source deleted	compute → REFUSE	claim_unavailable
source unreachable	compute → REFUSE	claim_unavailable
assumption text edited	compute → REFUSE	claim_changed

OBSERVED TRPB EVIDENCE, TYPED REUSE

Same context = ordinary cache. Of 488,601 recorded paid requests, 95.3% repeat an earlier one: typed reuse and a plain cache avoid exactly the same 465,564. Typed reuse gains nothing there.

Fault tests are not savings. In synthetic fixtures (4 injected fault types) a context-blind cache reused 2,564 invalid facts; typed reuse 0 and re-assayed them.

Bounded knowledge execution, not theorem extraction. Two existing claims, hand-authored adapters, recorded demo — not a live service. Adapters do **not** improve the TrpB route benchmark: a TrpB score is refused as a selection coefficient. Checks: 87 passed, 4 skipped in 2.27s; guard mutants 9/9, source mutants 7/7 killed.

Source `artifacts/claim_reuse/{demo_receipt,reuse_benchmark,verification_receipt}.json; interactive: site/claims/ · adapters reviewed by the authoring model, not a population geneticist (appendix) · artifacts @ a64c7210`

The contribution is compact: assay decisions with machine-checkable, conditional certificates and typed, reusable evidence.

CHECKED — CONDITIONALLY

Lean 4.35.0 compiled the generic route-soundness theorem, a weak "no certifiable route" cut and a strong cut; plus one 7-node VFVS → AIKG route recomputed exactly from raw CSV intervals (7/7); swapped-route mutant rejected (exit 1).

ASSUMED interval containment. Noise bins were fit on all replicates, so κ is a **transductive sensitivity**, not held-out validation. The 1,400 core and LazySP replay decisions are checked in Python; they are not individually kernel-checked.

PREDICTED — NOT MEASURED

Five BioNeMo ESM2 sizes, **8M–3B**, each scored all **160,000 variants** on Nebius H100s. Weak rank association did not improve with size. OpenResearch ran a companion analysis locally on CPU.

Replay using this prior: **1,100 follow-up episodes: no clear assay-saving benefit; all descriptive RMST intervals include zero**. Nothing here says it helps.

NEXT REAL EXPERIMENT

One lab. One prior campaign. One prospective round.

Pre-register the rule, equal assay budgets, read out cost and days to a defensible decision (120 days). No customer, LOI or market size exists.

Prior art: query-efficient path evaluation (LazySP adapted here; BAEE not run), ML-guided directed evolution (ALDE not run). **Still to establish:** a prospective wet-lab advantage. Model size and agent count did not establish a gain here.

Biologic design starts with justified enzyme experiments — one measured decision at a time.